ModelIt!

Student Activity Pack

Why Do You Look Like That?

The Genetic Lottery That Made You, You

Grade 8 | NGSS MS-LS3-1, MS-LS3-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you look similar to your parents but not identical?

2 What do you think determines your eye color?

3 Why might siblings look different from each other?

4 Draw what you think a 'gene' looks like and what it does.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Parent 1 Alleles | Parent 2 Alleles | Offspring Genotype | Trait Expression (Phenotype)

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Both Dominant Parents

BB × BB cross

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Carrier Cross

Bb × Bb cross

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Mixed Cross

Bb × bb cross

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Mutations — Random changes in DNA that create new alleles

Environmental Factors — Non-genetic influences that affect how traits are expressed

Multiple Genes — Many traits are controlled by more than one gene

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do you look similar to your parents but not exactly like them? Use evidence from your simulation to support your answers.

1 Why are you genetically unique even among siblings from the same parents?

2 How does sexual reproduction create more variation than asexual reproduction?

3 Why might genetic variation be important for a species' survival?

4 If two brown-eyed parents have a blue-eyed child, how is that possible?

5 How is a Punnett square similar to the model you built in ModelIt?

8 STEM Challenge: Solve a Genetic Mystery

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YOUR ENGINEERING MISSION

Use genetic inheritance patterns to solve a real-world mystery: determine the parents of an unknown organism based on trait evidence.

THE SCENARIO

A wildlife conservation team found a rare animal cub. They need to determine which adults are the parents to manage the breeding program. You have trait data for the cub and several potential parents.

Engineering Design Thinking

1. WHAT EVIDENCE WILL YOU USE TO IDENTIFY THE PARENTS?
2. HOW WILL YOU TEST YOUR HYPOTHESIS USING PUNNETT SQUARES?
3. CAN YOU IDENTIFY THE PARENTS WITH CERTAINTY, OR ONLY PROBABILITY?
4. WHAT ADDITIONAL TESTS MIGHT CONFIRM YOUR CONCLUSION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Genetic Counselors help families understand inherited conditions and make informed decisions. Forensic Geneticists use DNA to solve crimes. They earn $65,000-$110,000/year.